Theory Lectures on

Manufacturing Processes (MEI102)

AWJM – Abrasive Water Jet Machining

(Lecture 5)

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Machining – Subtractive Manufacturing

Seven different types of manufacturing processes

Casting | Forming | Joining | Machining | Powder Metallurgy | Additive Manufacturing | Surface Eng.

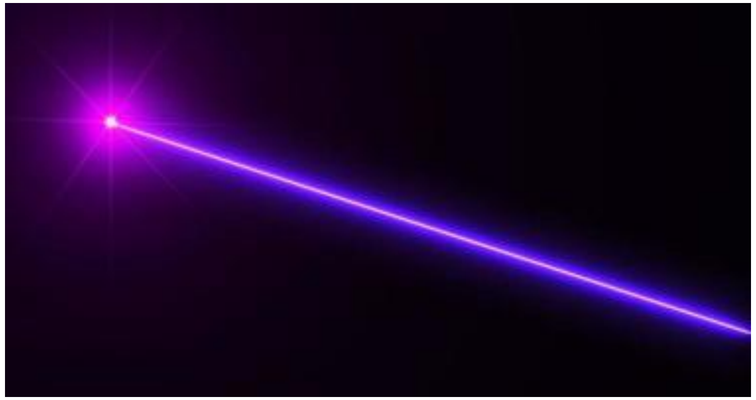
Machining OR Subtractive manufacturing

Traditional machining

Utilizes cutting tool
Based on mechanical energy

Non-traditional machining (NTM)

No special cutting tool
Mechanical/electrical/chemical/thermal/light etc. energy

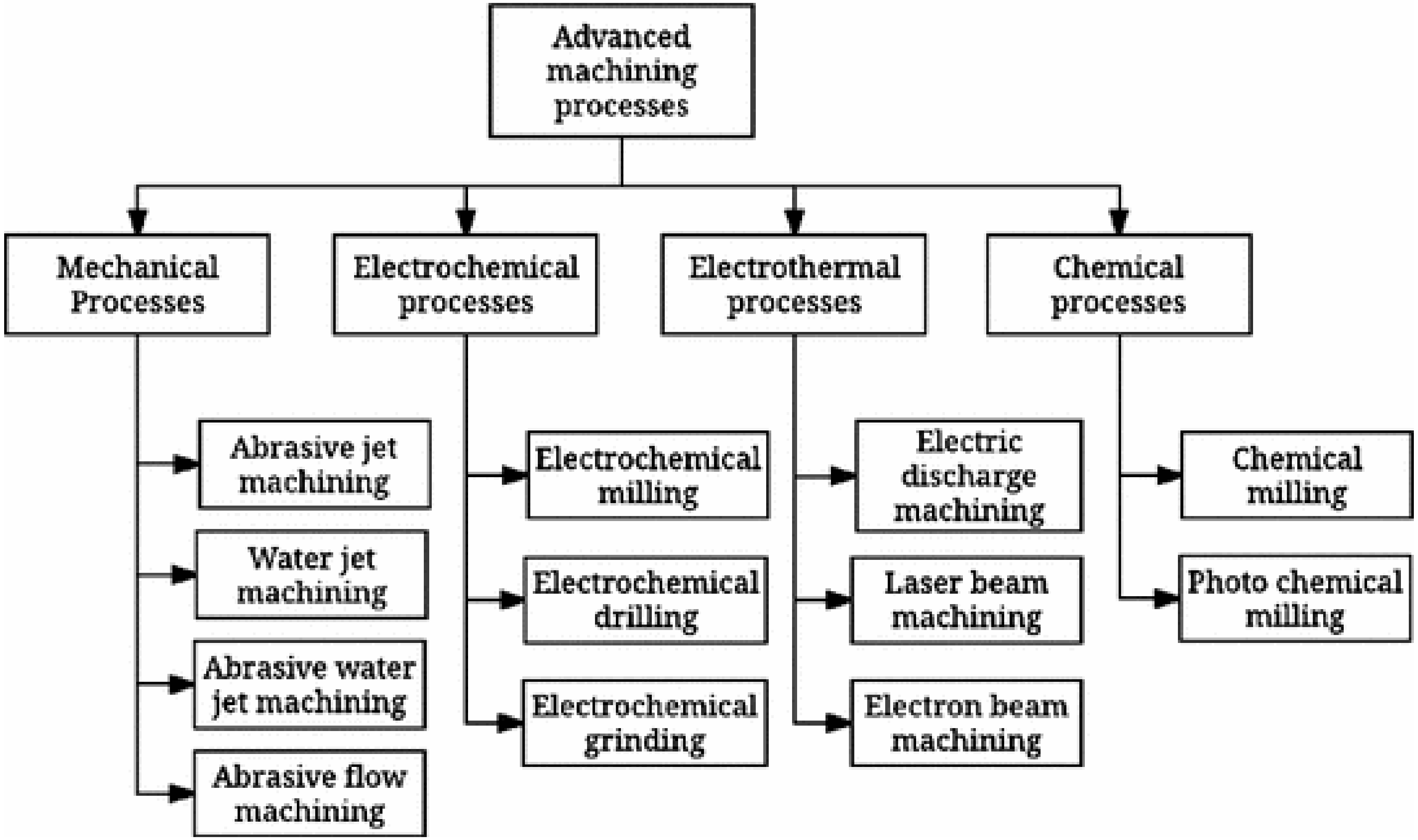


Both are "**subtractive**" because the material is removed from the initial workpiece in order to get the desired product.

Traditional and Non-traditional machining - difference

Traditional machining Conventional machining	Non-traditional machining Non-conventional machining
Only mechanical energy (power) is utilized to gradually remove excess material from workpiece.	**Various forms of energy** (like electrical, mechanical, thermal, chemical, electro-chemical, light, etc. or a combination of two or more such forms) are directly utilized to remove excess material.
Materials are removed in the form of tiny sharp **chips**.	So called 'chip' is not produced here. Material is removed in various forms, such as **tiny metal particles, ions, molten or vapor**, etc.
Physical contact between cutting tool and workpiece and also relative velocity in between them are indispensably necessary in order to remove materials.	Although no physical contact occurs between cutting tool and workpiece, in few processes such as AJM, AWJM, & USM solid abrasive grits strike the work surface to erode material.
'Shear deformation' is the only phenomenon that causes removal of material.	Shear deformation plays an insignificant role in material removing in most of the NTM processes. Instead, erosion, dissolution, evaporation, sputtering, etc. occur here.
Examples of conventional machining processes— Turning, Facing, Grooving, etc.Drilling, Boring, Reaming, etc.Shaping, Planing, Milling, etc.Grinding, Lapping, Honing, etc.	Examples of non-traditional machining processes— AJM, AWJM, USM, EDM, ECM, CHM, LBM, PAM, EBM, IBM, Hybridization of above processes.

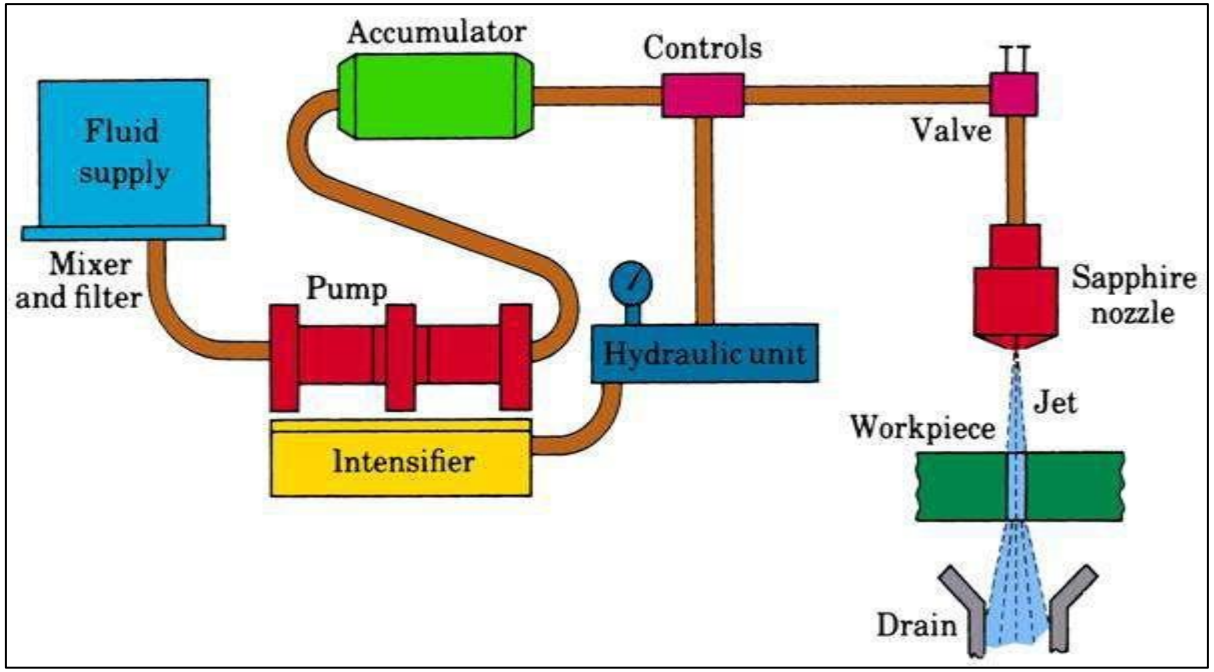
Classification of Non-traditional machining processes

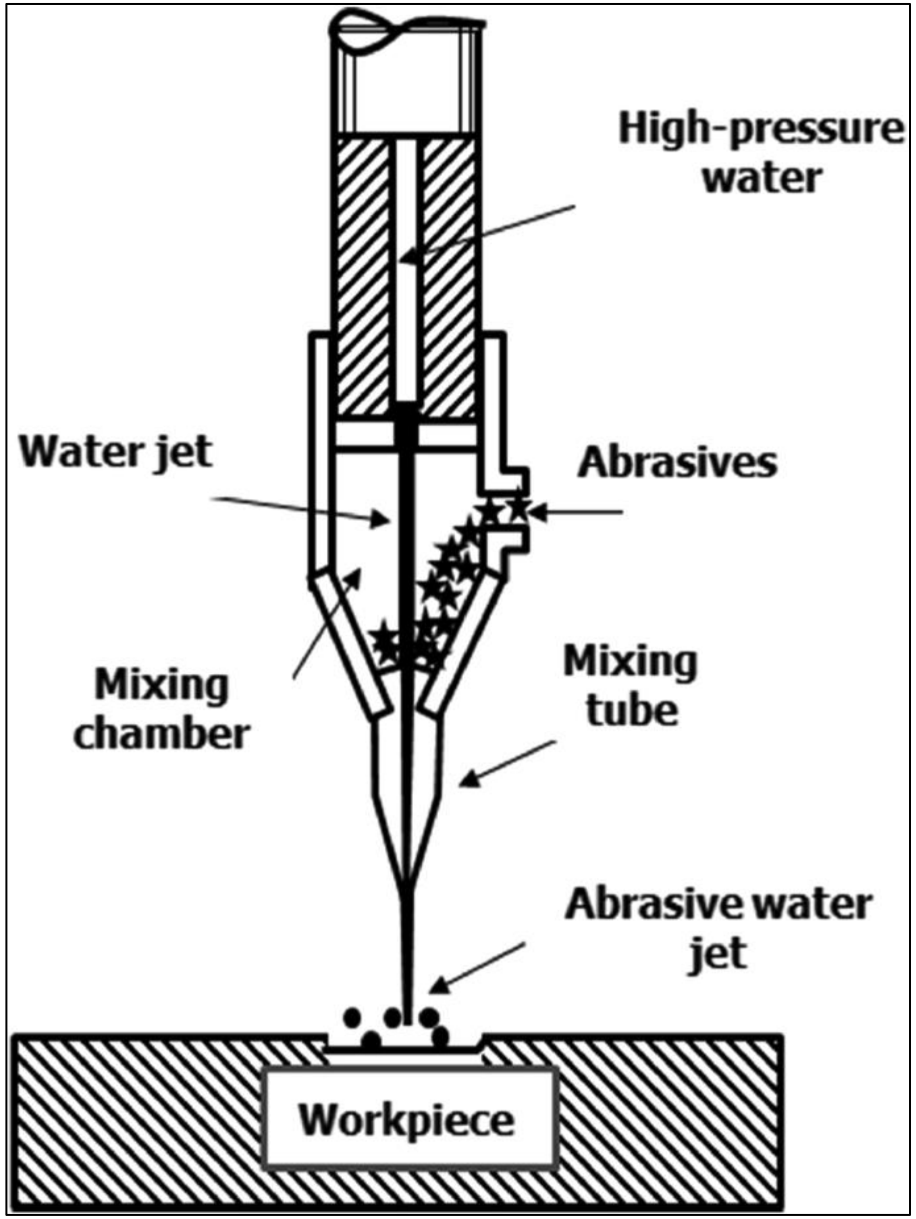


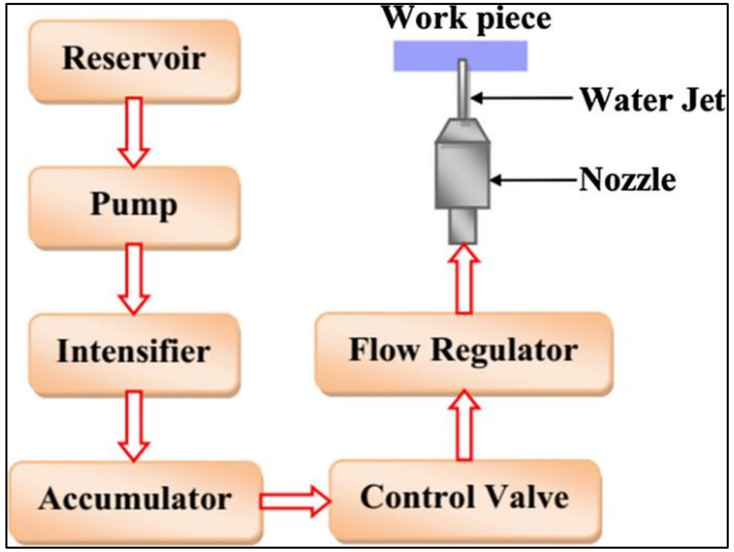
AWJM videos

1. https://www.youtube.com/shorts/Wvh_BGTNS3o
2. https://www.youtube.com/shorts/u8TwzsVvq6I
3. https://www.youtube.com/shorts/gqvWsXbIDz8
4. https://www.youtube.com/watch?v=EqpdPc7urGQ
5. https://www.youtube.com/watch?v=YlacOX68OME
6. https://www.youtube.com/watch?v=lQyWHLOxk5Y
7. https://www.youtube.com/watch?v=MVV17L4EbZM
8. https://www.youtube.com/watch?v=7iVHChvA7V0
9. https://www.youtube.com/watch?v=pemgwRrCs78
10. https://www.youtube.com/watch?v=JJV2tO7Ad1U

Abrasive Water Jet Machining (AWJM)



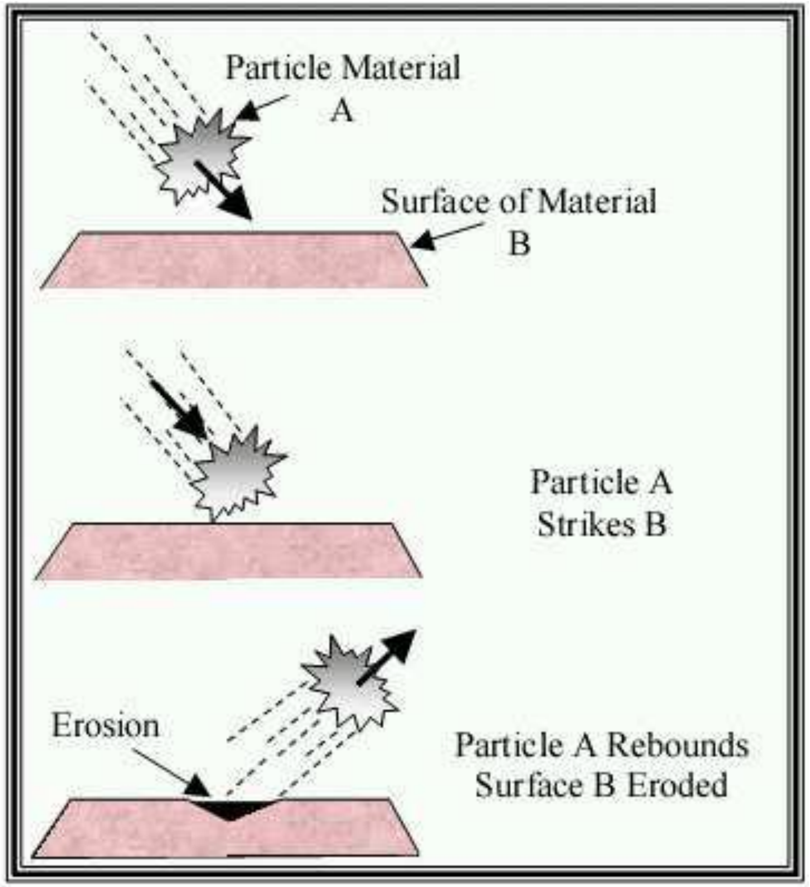




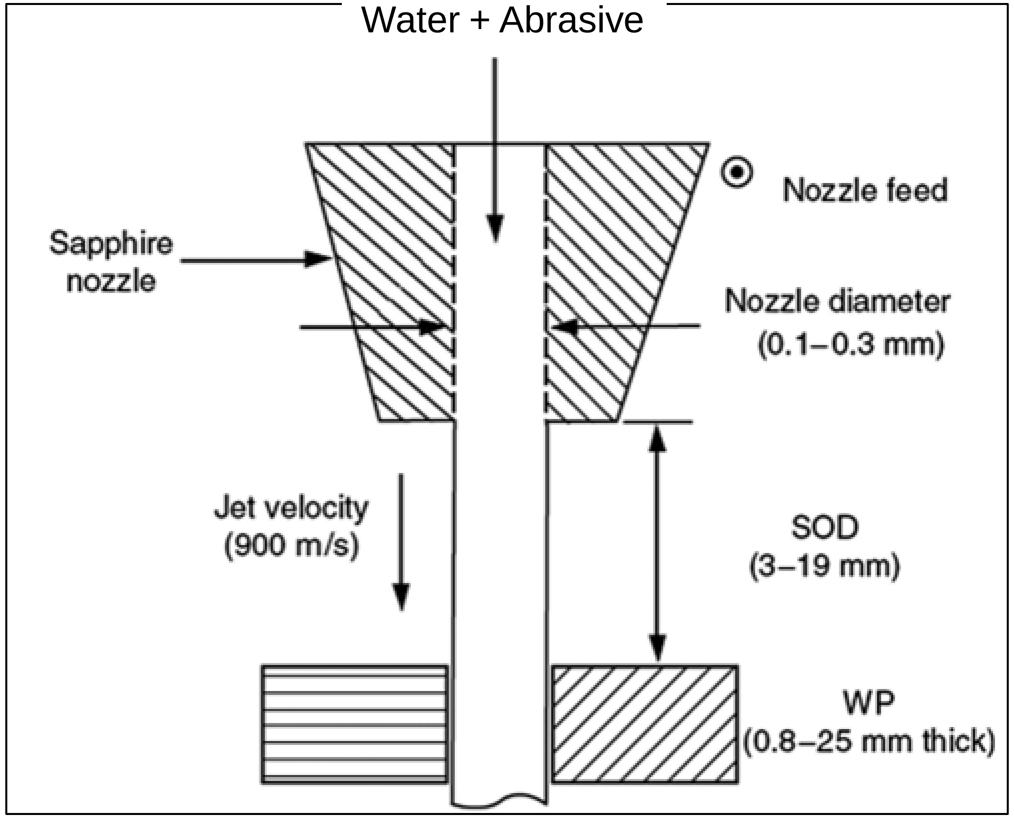
Mechanism of material removal in AWJM

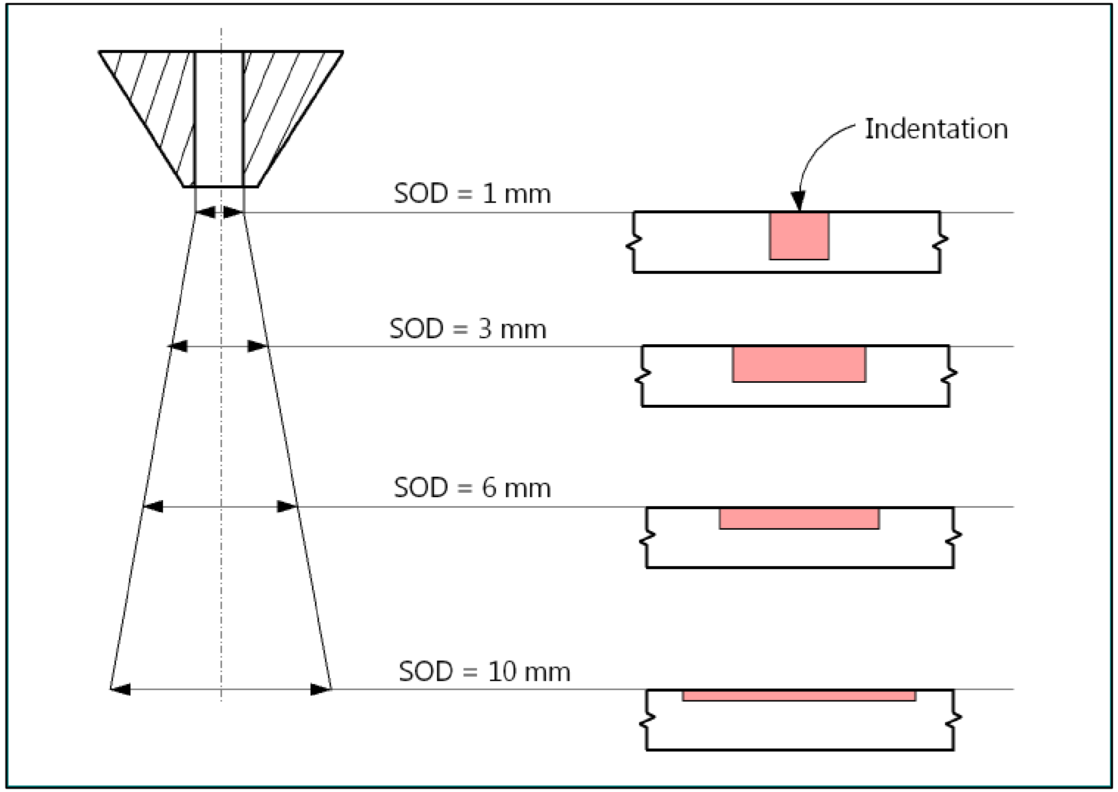
Salient features of AWJM

- AWJM comes under the mechanical-type non-traditional manufacturing process. (Mechanical energy based NTM process)
- The mechanism of material removal is ***abrasive erosion***.
- Small diameter hard abrasive particles are entrained into the highly pressurized water jet.
- The water-abrasive mixture is then dispensed through a small diameter nozzle in the form of a jet.
- This jet is directed to the workpiece that needs to be cut.
- The high-velocity abrasive particles induce erosion action on the material, whereas the eroded material is removed by the water. Therefore, the cutting action is executed by the abrasive particles, whereas water assists in removing the debris.



Stand-Off Distance (SOD)





- Abrasive-water jet start flaring immediately after it is dispensed from nozzle. As a result, cross-sectional area of the jet increases as the SOD (stand-off distance) increases. Jet cross-section is proportional to the SOD.
- Abrasive-water jet velocity (jet energy) is inversely proportional to the SOD.
- So, a small SOD gives narrow but deep profiles.
- On the other hand, a long SOD gives wide but shallow profile.

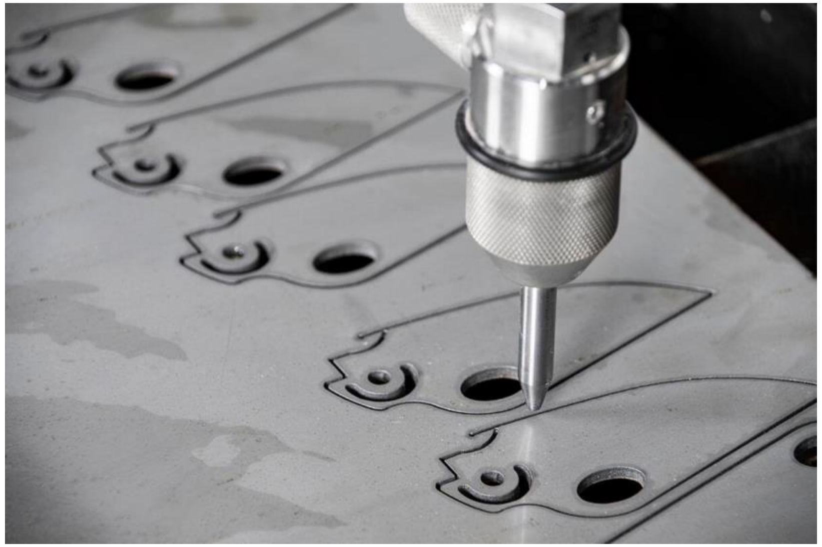
Common abrasive materials
- Silicon carbide
- Sand
- Corundum (alumina)
- Glass beads

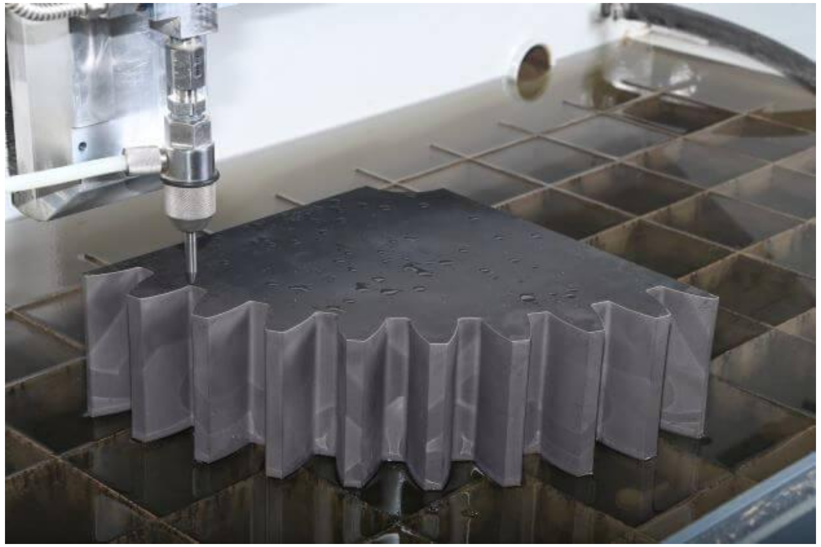
Typical abrasive size => 10 – 150 micron

- Nozzle diameter => 1 – 5 mm
- Nozzle material => Usually Sapphire
- Water pressure => 2,500 – 4,000 bar
- Abrasive materials => Silica, glass, alumina, etc.
- Abrasive flow rate => 0.1 – 1.0 kg/min
- SOD => 1 – 5 mm
- Impact angle => 60 – 90° with work surface
- Traverse speed => 100 – 5000 mm/min
- Depth of cut => 1.0 – 25.0 mm

Applications of AWJM

- Thick metal/ceramic cutting & profiling.
- Burr removal (deburring).
- Industrial and civil cleaning.
- Joint cleaning (rust removal from nut-bolt).
- Surface preparation and decontamination.
- Oil, paint, coating, etc. removal.
- Concrete hydro-demolition (stone cutting).
- Rock fragmentation.
- Soil drilling, rock drilling.
- Construction demolition.





Usually not used for cutting polymer or food product due to risk of presence of embedded abrasive particle

- Material Removal Rate.

$$MRR = u \times c_d \times \frac{\Pi}{4} d_o^{\,2} \times \sqrt{\frac{2 p_w^{\,3}}{\rho_w}}$$

Where

u	-	Constant that depends on the work material
c_d	-	Discharge coefficient of the orifice
d_o	-	diameter of the orifice
p_w	-	Pressure of water
ρ_w	-	Density of water

Problem 1:
During AWJM, if diameter of the nozzle orifice is reduced by 30%, how much decrease in MRR is expected?

Problem 2:
If hot water having density 0.8 times cold water is used in AWJM then how much reduction in MRR will be noticed?

Catcher for AWJM

- Even after completion of cutting action, the water jet can have sufficient energy that can damage machine elements if allowed to impinge the machine structure.

- Thus, the exit-jet also needs to be arrested. This is carried out by catcher. Catcher absorbs the residue energy of the exit-jet.
- Usually, catcher is based on either of these three different types.